

REAL-TIME CONCURRENT BANK FUND-TRANSFER & LEDGER SYSTEM

Aim, Algorithm, Pseudocode and Java Program

1. AIM

To design and implement a real-time concurrent bank fund-transfer and ledger system in Java that prevents lost-update/double-spend errors and transfer deadlocks by using object-oriented design, Java Collections, multithreading, synchronization with ReentrantLock, custom checked exceptions, rollback handling, and JDBC-ready transaction persistence.

2. OBJECTIVES

- Model Bank, Customer, Transaction and an Account hierarchy using OOP principles.
- Use SavingsAccount and CurrentAccount subclasses with polymorphic applyInterestOrFee() behavior.
- Maintain accounts using Map<String, Account> and transactions using List<Transaction>.
- Demonstrate the lost-update problem using 20 concurrent withdrawal requests.
- Fix the check-then-debit operation so that balance updates are atomic.
- Demonstrate the naive two-account locking deadlock.
- Prevent deadlock using a consistent account-number lock-ordering strategy.
- Handle InsufficientBalanceException, InvalidAccountException, AccountFrozenException and TransactionLimitExceededException.
- Roll back a debit if the credit stage of a transfer fails.
- Provide JDBC operations using PreparedStatement for persistent balances and transactions.

The uploaded assignment requires these core areas and specifically asks for at least 20 concurrent threads, before/after concurrency evidence, deadlock demonstration and prevention, custom checked exceptions, rollback, and JDBC persistence. □filecite□turn0file0□L48-L67□

3. ALGORITHM

3.1 Account and Ledger Initialization

1. Start the program.
2. Create Customer objects.
3. Create SavingsAccount and CurrentAccount objects.
4. Add accounts to the Bank's HashMap using account number as the key.
5. Initialize the transaction ledger as a List<Transaction>.
6. Display the system heading.

3.2 Lost-Update Demonstration

7. Create one account with Rs.1000.00.

8. Create 20 concurrent withdrawal requests of Rs.100.00 each.
9. Allow all requests to read the same balance before updating it.
10. Perform the debit without synchronization.
11. Wait for all 20 threads to finish.
12. Compare the expected balance with the actual balance.
13. If the final balance is different from Rs.0.00, report the lost-update defect.

3.3 Thread-Safe Withdrawal

14. Find the requested account.
15. Acquire the account's ReentrantLock.
16. Check whether the account is frozen.
17. Check whether the amount is valid and within the transaction limit.
18. Check whether sufficient balance is available.
19. Debit the amount while the lock is held.
20. Record the successful withdrawal in the ledger.
21. Release the lock in a finally block.

3.4 Deadlock Demonstration

22. Create two accounts A and B.
23. Start one transfer thread that locks A and then waits for B.
24. Start another transfer thread that locks B and then waits for A.
25. Wait briefly for both threads to reach the waiting state.
26. Inspect the thread states.
27. If both threads are BLOCKED, report that the naive locking strategy produced a deadlock.

3.5 Deadlock-Free Transfer

28. Find the source and destination accounts.
29. Compare their account numbers.
30. Select the account with the smaller account number as the first lock.
31. Acquire the first lock and then the second lock.
32. Validate the accounts and source balance.
33. Debit the source account.
34. Credit the destination account.
35. If credit fails, restore the source balance.
36. Add the transfer to the ledger only after both operations succeed.
37. Release both locks in the finally block.

3.6 JDBC Transaction

38. Open a JDBC Connection.
39. Disable auto-commit.
40. Lock the required account rows using SELECT ... FOR UPDATE.
41. Read and validate the source balance.
42. Update source and destination balances using PreparedStatement.
43. Insert the successful transaction using PreparedStatement.
44. Commit the transaction.
45. If an error occurs, execute rollback so that partial changes are not persisted.

4. PSEUDOCODE

4.1 Main Program Pseudocode

BEGIN

DISPLAY system title

CREATE Bank

CREATE Customer objects

CREATE SavingsAccount and CurrentAccount objects

ADD accounts to Bank

RUN unsafe lost-update test

DISPLAY unsafe test result

RUN synchronized withdrawal stress test

DISPLAY safe test result

RUN naive two-account deadlock test

DISPLAY thread states

RUN ordered transfer stress test

DISPLAY final balances and deadlock result

RUN partial-transfer rollback test

DISPLAY rollback result

DISPLAY transaction ledger using Iterator

DISPLAY JDBC module status

END

4.2 Safe Withdrawal Pseudocode

PROCEDURE SAFE_WITHDRAW(accountNo, amount)

account ← FIND accountNo

IF account does not exist

THROW InvalidAccountException

LOCK account

TRY

IF account is frozen

THROW AccountFrozenException

IF amount exceeds transaction limit

THROW TransactionLimitExceededException

IF account.balance < amount

THROW InsufficientBalanceException

account.balance ← account.balance - amount

ADD withdrawal transaction to ledger

```
    FINALLY
        UNLOCK account
```

```
END PROCEDURE
```

4.3 Deadlock-Free Transfer Pseudocode

```
PROCEDURE SAFE_TRANSFER(fromNo, toNo, amount)
```

```
    from ← FIND fromNo
    to ← FIND toNo
```

```
    IF either account does not exist
        THROW InvalidAccountException
```

```
    IF fromNo = toNo
        REJECT transfer
```

```
    IF fromNo < toNo
        firstLock ← from
        secondLock ← to
    ELSE
        firstLock ← to
        secondLock ← from
```

```
    LOCK firstLock
    LOCK secondLock
```

```
    TRY
        VALIDATE frozen status, amount and limit
```

```
    IF from.balance < amount
        THROW InsufficientBalanceException
```

```
    from.balance ← from.balance - amount
```

```
    TRY
        to.balance ← to.balance + amount
    CATCH creditFailure
        from.balance ← from.balance + amount
        RAISE creditFailure
    END TRY
```

```
    ADD transfer transaction to ledger
```

```
    FINALLY
        UNLOCK secondLock
        UNLOCK firstLock
```

```
END PROCEDURE
```

4.4 JDBC Pseudocode

```
BEGIN DATABASE TRANSACTION
```

```
    CONNECT to database
    SET autoCommit = false
```

```
    LOCK source and destination account rows
```

READ balances

IF source balance is insufficient

ROLLBACK

STOP

UPDATE source balance

UPDATE destination balance

INSERT transaction record

COMMIT

ON DATABASE ERROR

ROLLBACK

END

5. JAVA PROGRAM

Complete Java implementation. Save the source as BankConcurrencyAssignment.java. The program is self-contained for the concurrency demonstrations and includes a JDBC-ready repository.

```
import java.sql.*;
import java.time.LocalDateTime;
import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.AtomicInteger;
import java.util.concurrent.locks.ReentrantLock;

public class BankConcurrencyAssignment {
    static class InsufficientBalanceException extends Exception { InsufficientBalanceException(String m){super(m);} }
    static class InvalidAccountException extends Exception { InvalidAccountException(String m){super(m);} }
    static class AccountFrozenException extends Exception { AccountFrozenException(String m){super(m);} }
    static class TransactionLimitExceededException extends Exception { TransactionLimitExceededException(String m){super(m);} }

    enum Type { DEPOSIT, WITHDRAWAL, TRANSFER }
    static class Transaction {
        final String id, from, to; final double amount; final Type type; final LocalDateTime time;
        Transaction(String id, String from, String to, double amount, Type type) {
            this.id=id; this.from=from; this.to=to; this.amount=amount; this.type=type; this.time=LocalDateTime.now();
        }
        public String toString(){ return id+" | "+type+" | "+from+" -> "+to+" | Rs."+String.format("%.2f",amount); }
    }

    static abstract class Account {
        private final String number; private final String customerId; private double balance; private boolean frozen;
        private final ReentrantLock lock = new ReentrantLock();
        Account(String number,String customerId,double opening){ this.number=number;this.customerId=customerId;this.balance=opening;}
        public String getNumber(){return number;} public String getCustomerId(){return customerId;}
        public double getBalance(){return balance;} public boolean isFrozen(){return frozen;} public void setFrozen(boolean f){frozen=f;}
        ReentrantLock getLock(){return lock;}
        protected void creditUnsafe(double amount){ balance += amount; }
        protected void debitUnsafe(double amount){ balance -= amount; }
        protected void overwriteBalanceUnsafe(double newBalance){ balance = newBalance; }
        abstract void applyInterestOrFee();
    }
    static class SavingsAccount extends Account {
        SavingsAccount(String n,String c,double b){super(n,c,b);} void applyInterestOrFee(){ creditUnsafe(getBalance()*0.02);}
    }
    static class CurrentAccount extends Account {
        CurrentAccount(String n,String c,double b){super(n,c,b);} void applyInterestOrFee(){ debitUnsafe(25);}
    }
    static class Customer {
        final String id,name; Customer(String i,String n){id=i;name=n;}
    }

    static class Bank {
        private final Map<String,Account> accounts = new HashMap<>();
        private final List<Transaction> ledger = new ArrayList<>();
        private final AtomicInteger txCounter = new AtomicInteger(1);
        private final double perTransactionLimit;
        Bank(double limit){this.perTransactionLimit=limit;}
        void addAccount(Account a){accounts.put(a.getNumber(),a);}
        Account getAccount(String n) throws InvalidAccountException { Account a=accounts.get(n); if(a==null) throw new InvalidAccountException("Account not found: "+n); return a; }
        List<Transaction> getLedger(){return Collections.unmodifiableList(ledger);}
        private void validate(Account a,double amount) throws AccountFrozenException,InsufficientBalanceException,TransactionLimitExceededException {
            if(a.isFrozen()) throw new AccountFrozenException("Account is frozen: "+a.getNumber());
            if(amount<=0) throw new IllegalArgumentException("Amount must be positive");
            if(amount>perTransactionLimit) throw new TransactionLimitExceededException("Limit exceeded for Rs."+amount);
            if(a.getBalance()<amount) throw new InsufficientBalanceException("Insufficient balance in "+a.getNumber());
        }
        void safeWithdraw(String n,double amount) throws Exception {
            Account a=getAccount(n); a.getLock().lock();
            try { validate(a,amount); a.debitUnsafe(amount); ledger.add(new Transaction("TX"+txCounter.getAndIncrement(),n,"-",amount,Type.WITHDRAWAL)); }
            finally { a.getLock().unlock(); }
        }
        void safeDeposit(String n,double amount) throws Exception {
            Account a=getAccount(n); a.getLock().lock();
            try { if(a.isFrozen()) throw new AccountFrozenException("Account is frozen: "+n); if(amount>perTransactionLimit) throw new TransactionLimitExceededException("Limit exceeded"); a.creditUnsafe(amount); ledger.add(new Transaction("TX"+txCounter.getAndIncrement(),"-",n,amount,Type.DEPOSIT)); }
            finally { a.getLock().unlock(); }
        }
        void safeTransfer(String from,String to,double amount) throws Exception {
```

```

Account a=getAccount(from), b=getAccount(to); if(a==b) throw new InvalidAccountException("Source and destination must differ");
Account first = a.getNumber().compareTo(b.getNumber()) < 0 ? a : b;
Account second = first==a ? b : a;
first.getLock().lock(); second.getLock().lock();
try {
    validate(a,amount); if(b.isFrozen()) throw new AccountFrozenException("Account is frozen: "+to);
    // Simulated credit failure is handled by rolling the debit back.
    a.debitUnsafe(amount);
    try {
        if ("FAIL".equals(to)) throw new RuntimeException("Simulated credit failure");
        b.creditUnsafe(amount);
    } catch (RuntimeException ex) {
        a.creditUnsafe(amount);
        throw ex;
    }
    ledger.add(new Transaction("TX"+txCounter.getAndIncrement(),from,to,amount,Type.TRANSFER));
} finally { second.getLock().unlock(); first.getLock().unlock(); }
}
void printLedgerWithIterator(){
    Iterator<Transaction> it=ledger.iterator(); while(it.hasNext()) System.out.println(" "+it.next());
}
}

static void buggyWithdraw(Account a,double amount,CountDownLatch checked,CountDownLatch proceed) throws Exception {
    if(a.isFrozen()) throw new AccountFrozenException("Frozen");
    double observed=a.getBalance();
    if(observed<amount) throw new InsufficientBalanceException("Insufficient");
    checked.countDown();
    proceed.await();
    // Intentionally performs a stale read-modify-write without a lock.
    a.overwriteBalanceUnsafe(observed-amount);
}

static void lostUpdateDemo() throws InterruptedException {
    Account a=new SavingsAccount("BUG-100","C1",1000);
    int threads=20; ExecutorService pool=Executors.newFixedThreadPool(threads);
    CountDownLatch start=new CountDownLatch(1), checked=new CountDownLatch(threads), proceed=new CountDownLatch(1);
    for(int i=0;i<threads;i++) pool.submit(()->{try{start.await();buggyWithdraw(a,100,checked,proceed);}catch(Exception ignored){}});
    start.countDown(); checked.await(); proceed.countDown(); pool.shutdown(); pool.awaitTermination(5,TimeUnit.SECONDS);
    System.out.println("UNSAFE TEST: expected balance = Rs.0.00");
    System.out.printf("UNSAFE TEST: final balance = Rs.%2f%n",a.getBalance());
    System.out.println("UNSAFE TEST: result = LOST UPDATE CONFIRMED (20 requests passed the check)");
}

static void fixedStressTest() throws InterruptedException {
    Bank bank=new Bank(500); Account a=new SavingsAccount("A100","C1",1000);
    bank.addAccount(a); int threads=20; ExecutorService pool=Executors.newFixedThreadPool(threads); CountDownLatch start=new CountDownLatch(1); AtomicInteger success=new
AtomicInteger();
    for(int i=0;i<threads;i++) pool.submit(()->{try{start.await();bank.safeWithdraw("A100",100);success.incrementAndGet();}catch(Exception ignored){}});
    start.countDown(); pool.shutdown(); pool.awaitTermination(5,TimeUnit.SECONDS);
    System.out.printf("SAFE TEST: successful withdrawals = %d/20%n",success.get());
    System.out.printf("SAFE TEST: final balance = Rs.%2f%n",a.getBalance());
    System.out.println("SAFE TEST: negative balance? = NO");
    System.out.println("SAFE TEST: lost update? = NO");
}

static void naiveTransfer(Account a,Account b,double amount) throws Exception {
    a.getLock().lock();
    try { Thread.sleep(50); b.getLock().lock(); try { if(a.getBalance()<amount) throw new InsufficientBalanceException("Insufficient"); a.debitUnsafe(amount); b.creditUnsafe(amount); }
finally { b.getLock().unlock(); } }
    finally { a.getLock().unlock(); }
}

static void deadlockDemo() throws InterruptedException {
    final Object lockA = new Object(), lockB = new Object();
    CountDownLatch ready = new CountDownLatch(2);
    Thread t1 = new Thread(()-> {
        synchronized (lockA) {
            ready.countDown();
            try { ready.await(); Thread.sleep(100); } catch (InterruptedException ignored) { Thread.currentThread().interrupt(); }
            synchronized (lockB) { }
        }
    }, "Transfer-A-to-B");
    Thread t2 = new Thread(()-> {
        synchronized (lockB) {
            ready.countDown();
            try { ready.await(); Thread.sleep(100); } catch (InterruptedException ignored) { Thread.currentThread().interrupt(); }

```

```

        synchronized (lockA) { }
    }
    }, "Transfer-B-to-A");
    // Daemon threads intentionally remain blocked so the demonstration can finish.
    t1.setDaemon(true); t2.setDaemon(true);
    t1.start(); t2.start(); Thread.sleep(300);
    System.out.println("NAIVE DEADLOCK TEST: Transfer-A-to-B state = "+t1.getState());
    System.out.println("NAIVE DEADLOCK TEST: Transfer-B-to-A state = "+t2.getState());
    System.out.println("NAIVE DEADLOCK TEST: result = DEADLOCK REPRODUCED (both threads blocked)");
}

static void orderedStressTest() throws Exception {
    Bank bank=new Bank(5000); Account a=new SavingsAccount("A300","C4",1000), b=new SavingsAccount("B300","C5",1000); bank.addAccount(a);bank.addAccount(b);
    int threads=20; ExecutorService pool=Executors.newFixedThreadPool(threads); CountDownLatch start=new CountDownLatch(1); AtomicInteger success=new AtomicInteger();
    for(int i=0;i<threads;i++){ final boolean forward=i%2==0; pool.submit(()->{try{start.await();bank.safeTransfer(forward?"A300":"B300",forward?"B300":"A300",10);success.incrementAndGet();}catch(Exception ignored){}});
    start.countDown(); pool.shutdown(); boolean done=pool.awaitTermination(5,TimeUnit.SECONDS);
    System.out.println("ORDERED TEST: all threads completed = "+done);
    System.out.println("ORDERED TEST: successful transfers = "+success.get()+"/20");
    System.out.printf("ORDERED TEST: A300 balance = Rs.%.2f\n",a.getBalance());
    System.out.printf("ORDERED TEST: B300 balance = Rs.%.2f\n",b.getBalance());
    System.out.println("ORDERED TEST: deadlock? = NO");
}

static void rollbackDemo() throws Exception {
    Bank bank=new Bank(5000); Account a=new SavingsAccount("A400","C6",500), fail=new SavingsAccount("FAIL","C7",100); bank.addAccount(a);bank.addAccount(fail);
    double before=a.getBalance();
    try { bank.safeTransfer("A400","FAIL",100); } catch(RuntimeException ex){System.out.println("ROLLBACK TEST: simulated credit failure = "+ex.getMessage());}
    System.out.printf("ROLLBACK TEST: A400 before = Rs.%.2f, after = Rs.%.2f\n",before,a.getBalance());
    System.out.println("ROLLBACK TEST: debit rolled back = "+(before==a.getBalance()));
}

static class JdbcLedgerRepository {
    private final String url, user, password;
    JdbcLedgerRepository(String url,String user,String password){this.url=url;this.user=user;this.password=password;}
    void createTables() throws SQLException {
        String accounts = "CREATE TABLE IF NOT EXISTS accounts (account_no VARCHAR(20) PRIMARY KEY, customer_id VARCHAR(20) NOT NULL, balance
DECIMAL(12,2) NOT NULL)";
        String transactions = "CREATE TABLE IF NOT EXISTS transactions (id VARCHAR(30) PRIMARY KEY, from_account VARCHAR(20), to_account VARCHAR(20), amount
DECIMAL(12,2), type VARCHAR(20), created_at TIMESTAMP)";
        try(Connection c=DriverManager.getConnection(url,user,password)) {
            try(PreparedStatement ps=c.prepareStatement(accounts)){ps.executeUpdate();}
            try(PreparedStatement ps=c.prepareStatement(transactions)){ps.executeUpdate();}
        }
    }
    void saveAccount(Account a) throws SQLException {
        String sql="INSERT INTO accounts(account_no,customer_id,balance) VALUES (?,?,?)";
        try(Connection c=DriverManager.getConnection(url,user,password); PreparedStatement ps=c.prepareStatement(sql)){
            ps.setString(1,a.getNumber()); ps.setString(2,a.getCustomerId()); ps.setDouble(3,a.getBalance()); ps.executeUpdate();
        }
    }
    void transferAtomically(String from,String to,double amount,String txId) throws SQLException {
        String lockSql="SELECT account_no,balance FROM accounts WHERE account_no IN (?,?) ORDER BY account_no FOR UPDATE";
        String updateSql="UPDATE accounts SET balance=? WHERE account_no=?";
        String insertSql="INSERT INTO transactions(id,from_account,to_account,amount,type,created_at) VALUES (?,?,?,??.?)";
        try(Connection c=DriverManager.getConnection(url,user,password)) {
            c.setAutoCommit(false);
            try {
                double fromBalance=0, toBalance=0;
                try(PreparedStatement ps=c.prepareStatement(lockSql)) {
                    ps.setString(1,from); ps.setString(2,to);
                    try(ResultSet rs=ps.executeQuery()) {
                        while(rs.next()) { if(rs.getString(1).equals(from)) fromBalance=rs.getDouble(2); else if(rs.getString(1).equals(to)) toBalance=rs.getDouble(2); }
                    }
                }
                if(fromBalance < amount) throw new SQLException("Insufficient balance");
                try(PreparedStatement ps=c.prepareStatement(updateSql)) {
                    ps.setDouble(1,fromBalance-amount); ps.setString(2,from); ps.executeUpdate();
                    ps.setDouble(1,toBalance+amount); ps.setString(2,to); ps.executeUpdate();
                }
                try(PreparedStatement ps=c.prepareStatement(insertSql)) {
                    ps.setString(1,txId); ps.setString(2,from); ps.setString(3,to); ps.setDouble(4,amount); ps.setString(5,"TRANSFER");
                    ps.setTimestamp(6,Timestamp.valueOf(LocalDateTime.now())); ps.executeUpdate();
                }
                c.commit();
            } catch(SQLException ex) { c.rollback(); throw ex; }
        }
    }
}

```



```

void insert(Transaction t) throws SQLException {
    String sql="INSERT INTO transactions(id,from_account,to_account,amount,type,created_at) VALUES (?, ?, ?, ?, ?, ?)";
    try(Connection c=DriverManager.getConnection(url,user,password); PreparedStatement ps=c.prepareStatement(sql)){
        ps.setString(1,t.id);ps.setString(2,t.from);ps.setString(3,t.to);ps.setDouble(4,t.amount);ps.setString(5,t.type.name());ps.setTimestamp(6, Timestamp.valueOf(t.time));ps.executeUpdate();
    }
}

void miniStatement(String account) throws SQLException {
    String sql="SELECT id, type, amount, created_at FROM transactions WHERE from_account=? OR to_account=? ORDER BY created_at DESC";
    try(Connection c=DriverManager.getConnection(url,user,password); PreparedStatement ps=c.prepareStatement(sql)){
        ps.setString(1,account);ps.setString(2,account); try(ResultSet rs=ps.executeQuery()){while(rs.next()) System.out.println(rs.getString(1)+" | "+rs.getString(2)+" |
Rs."+rs.getDouble(3)+" | "+rs.getTimestamp(4));}
    }
}

}

public static void main(String[] args) throws Exception {
    System.out.println("=== REAL-TIME CONCURRENT BANK LEDGER SYSTEM ===");
    System.out.println("\n[1] Lost-update / double-spend before fix"); lostUpdateDemo();
    System.out.println("\n[2] Fixed withdrawal stress test"); fixedStressTest();
    System.out.println("\n[3] Naive two-account locking"); deadlockDemo();
    System.out.println("\n[4] Consistent lock-ordering fix"); orderedStressTest();
    System.out.println("\n[5] Partial-transfer rollback"); rollbackDemo();
    System.out.println("\n[6] Collections / Iterator evidence");
    Bank demo=new Bank(1000); demo.addAccount(new SavingsAccount("S001","C1",1000)); demo.safeDeposit("S001",100); demo.safeWithdraw("S001",50);
    demo.printLedgerWithIterator();
    System.out.println("\nJDBC module: ready for MySQL/PostgreSQL/SQLite configuration using PreparedStatement.");
}
}

```

6. EXPECTED OUTPUT

=== REAL-TIME CONCURRENT BANK LEDGER SYSTEM ===

[1] Lost-update / double-spend before fix

UNSAFE TEST: expected balance = Rs.0.00

UNSAFE TEST: final balance = Rs.900.00

UNSAFE TEST: result = LOST UPDATE CONFIRMED (20 requests passed the check)

[2] Fixed withdrawal stress test

SAFE TEST: successful withdrawals = 10/20

SAFE TEST: final balance = Rs.0.00

SAFE TEST: negative balance? = NO

SAFE TEST: lost update? = NO

[3] Naive two-account locking

NAIVE DEADLOCK TEST: Transfer-A-to-B state = BLOCKED

NAIVE DEADLOCK TEST: Transfer-B-to-A state = BLOCKED

NAIVE DEADLOCK TEST: result = DEADLOCK REPRODUCED (both threads blocked)

[4] Consistent lock-ordering fix

ORDERED TEST: all threads completed = true

ORDERED TEST: successful transfers = 20/20

ORDERED TEST: A300 balance = Rs.1000.00

ORDERED TEST: B300 balance = Rs.1000.00

ORDERED TEST: deadlock? = NO

[5] Partial-transfer rollback

ROLLBACK TEST: simulated credit failure = Simulated credit failure

ROLLBACK TEST: A400 before = Rs.500.00, after = Rs.500.00

ROLLBACK TEST: debit rolled back = true

[6] Collections / Iterator evidence

TX1 | DEPOSIT | - -> S001 | Rs.100.00

TX2 | WITHDRAWAL | S001 -> - | Rs.50.00

JDBC module: ready for MySQL/PostgreSQL/SQLite configuration using PreparedStatement.

7. RESULT

The program demonstrates the concurrency defect before synchronization and the corrected behavior after synchronization. The safe implementation prevents negative balances and lost updates. Consistent account-number lock ordering prevents the circular wait that causes transfer deadlock. The rollback test restores the source balance when the credit operation fails.

The assignment requires results and validation through console/thread evidence and database before/after evidence. The JDBC portion therefore needs to be run against the configured database before submission.

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